

Question: In how many ways can 7 people sit around a round table if no two of the 3 people Pierre, Rosa, and Thomas can sit next to each other? (Seating arrangements which are rotations of each other are treated as the same.)

Answer:

SABER

<think>\nOkay, so I need to figure out how many ways 7 people can sit around a round table, but with the condition that Pierre, Rosa, and Thomas (let's call them P, R, T for short) can't sit next to each other.

<omitted>

So the number of ways would be $C(4,3) * 3! = 4 * 6 = 24$. \n\nWait

SC: 0.33 CS: 0.91 RSS : 0.73

, but hold on. Let me think again. If we fix the positions of the other 4 people, then there are 4 gaps. To place P, R, T into different gaps, we need to choose 3 gaps out of 4, and then arrange P, R, T in those gaps.

<omitted>

but since we fixed one person's position to account for circular permutations? \n\nWait

SC: 0.60 CS: 0.95 RSS : 0.84

, maybe I need to adjust for circular permutations. Let me clarify. \n\nFirst, when arranging people around a circular table, fixing one person's position is a common technique to convert it into a linear permutation problem. So if I fix one person, say, Pierre, then arrange the rest.

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So total number of ways would be $6 * 4 * 6 = 144$. But wait, is this correct? \n\nWait

SC: 1.0 CS: 0.99 RSS : 0.99>0.95

</think>

\n\nTo solve the problem of seating 7 people around a round table such that **Pierre (P), Rosa (R), and Thomas (T)** are **not adjacent to each other**, we proceed step by step using the **gap method** in circular permutations.

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$\times \binom{4}{3} \times 3! = 6 \times 4 \times 6 = 144$ \n\n--- \n\n### ☒ Final

Answer \n\n\$ \boxed{144} \n\n\$

Vanilla

<think>\nOkay, so I need to figure out how many ways 7 people can sit around a round table, but with the condition that Pierre, Rosa, and Thomas (let me call them P, R, T for short) can't sit next to each other. And rotations are considered the same, so it's a circular permutation problem with restrictions. \n\nFirst, I remember that for circular permutations, the number of ways to arrange n people is $(n-1)!$ because rotations are considered the same. So normally, without any restrictions, 7 people would have $6! = 720$ ways. But here, we have restrictions on P, R, and T. \n\nThe problem says no two of them can sit next to each other. So, I need to subtract the arrangements where at least two of them are sitting together. **But wait**, inclusion-exclusion might come into play here. Let me think. \n\nAlternatively, maybe it's easier to first arrange the other people and then place P, R, T in the gaps?

<omitted>

So the number of ways would be $C(4,3) * 3! = 4 * 6 = 24$. \n\nTherefore, total number of arrangements would be $3! * 4 * 3! = 6 * 24 = 144$.

\n\nWait, but hold on. Let me check again. \n\nFirst, arranging the 4 people around the table: $(4-1)! = 6$. Then, placing the 3 people into the gaps: choosing 3 gaps out of 4, which is $C(4,3) = 4$, and then arranging P, R, T in those gaps, which is $3! = 6$. So total is $6 * 4 * 6 = 144$.

\n\nBut wait, is that correct? Let me think again.

<omitted>

So the number of entities is 5: the trio and the other 4 people. So circular arrangements: $(5-1)! = 4! = 24$. Then, the trio can be arranged in $3!$ ways. So $|A \cap B| = 3! * 4! = 6 * 24 = 144$. \n\nWait, but hold on. Wait, if we have three people together, how do we count? Let me think.

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\n\nAs calculated, this is $2 * 5! = 240$. \n\nBut if we fix P's position, how many arrangements have P and R together? \n\nWith P fixed, R can be in seat 2 or seat 7 (adjacent to P). Then, the rest can be arranged in $5!$ ways? No, with P fixed, the remaining 6 seats are arranged, but if R is fixed next to P, then the remaining 5 people can be arranged in $5!$ ways.

\n\nWait, but with

Endless exhaustive search